1.3 Biomechanics

This topic will focus on the biomechanics of movement. It involves the study of force and its effect on human movement in physical activities and sports.

The ultimate goal of biomechanics is to improve performance and the prevention and treatment injury by optimising technique, training and equipment in physical activity and sport.

The study of biomechanical movement will allow learners to develop their knowledge and understanding of the more technical aspects of performance and participation in physical activity and sport and evaluate their own and others' effectiveness and efficiency.

This topic will develop learners' knowledge and understanding of biomechanical principles, including defining and applying Newton's Laws. The concept of force will be understood along with being able to draw and understand free body diagrams.

Learners will develop their knowledge and understanding of levers and the mechanical advantage of the second class lever, as well as the use of technology to analyse movement and improve performance.

The definitions and creation of linear motion and angular motion will be known and learners will be able to make calculations for quantities of linear and angular motion.

Learners will develop their knowledge and understanding of fluid mechanics, including factors that impact the magnitude of air resistance (on land) or drag (in water) on a body or object.

Learners will also develop their knowledge and understanding of projectile motion, including the application of Bernoulli's principle.